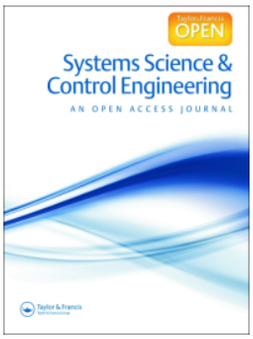

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RESEARCH ARTICLE

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Enhancing intrusion detection in IoT networks using machine learning-based feature selection and ensemble models

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ABSTRACT

Internet of Things (IoT) technology has evolved significantly, transitioning from personal devices to powering smart cities and global deployments across diverse industries. However, security challenges arise due to diverse devices using various protocols and having limited computational capabilities, leading to vulnerabilities and potential intrusions in IoT networks. This paper addresses the challenge of intrusion detection in IoT by introducing a heterogeneous machine learning-based stack classifier model for IoT data. The model employs feature selection and ensemble modelling to investigate and enhance key classification metrics for intrusion detection of IoT data. This approach comprises two core components: the utilization of the K-Best algorithm for feature selection, extracting the top 15 critical features and the construction of an ensemble model incorporating various traditional machine learning models. The integration of these components harnesses information from selected features and leverages the collective strength of individual models to enhance classification performance. Using the 'Ton IoT dataset,' our experiments compare the ensemble model with individual ones. This research aims to improve key classification metrics for IoT intrusion detection, focusing on accuracy, precision, recall and F1 score. Through rigorous experimentation and comparisons, the proposed ensemble approach showcases exceptional performance, providing a robust solution to fortify IoT network security.

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Internet of Things (IoT); intrusion detection; machine learning (ML); feature selection; ensemble learning

1. Introduction

Data security is of paramount importance in today's data-driven era, particularly with the exponential growth in data volume. This underscores the critical significance of information security, encompassing not only individualized documentation but also information owned and utilized by diverse institutions (Khashan & Khafajah, 2018). Safeguarding the confidentiality, integrity, and availability of data is essential to prevent unauthorized access, alteration, or deletion and to maintain trust among various individual users and the public (Khashan et al., 2023c).

The proliferation of the Internet of Things (IoT) has introduced heightened vulnerabilities in various services and systems, making them targets for external attacks aiming to acquire sensitive personal data. Intrusions in both centrally connected and decentralized IoT networks have surged due to expanding connectivity (Khashan et al., 2021b). Research has explored into the multifaceted role of IoT across various domains and the effectiveness of intrusion detection systems (IDS) in countering these intrusions (Mashal et al., 2023). As the need for robust intrusion detection solutions grows, machine learning technologies have emerged as pivotal tools in achieving early intrusion detection (Khan & Javaid, 2022).

However, IoT networks pose distinctive security challenges due to the diverse range of interconnected devices with varying protocols and computational capabilities. Traditional IDS, designed for standardized and centralized networks, face limitations in adapting to the dynamic and heterogeneous IoT environment. The lack of standardization in IoT security measures, coupled with resource constraints on IoT devices, hinders the effectiveness of traditional intrusion detection methods. Additionally, the decentralized nature of IoT architectures complicates security management for traditional systems. Therefore, there is a clear need for innovative intrusion detection approaches tailored to the specific challenges of IoT networks (Thakkar & Lohiya, 2021).

IoT constitutes a network of interconnected devices generating data within the network, relying on real-time communication among sensor-equipped devices (Da Costa et al., 2019) Across industries, IoT facilitates

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data-driven decision-making by providing real-time monitoring and data on routine activities, granting a competitive advantage to various stakeholders and leaders (Khashan, 2020b). The seamless exchange of information across corporate operations also significantly influences overall business performance (Khashan, 2020a). IoT networks are susceptible to hacker attacks due to weak security credentials, inadequate firmware vulnerability assessments, and infrequent security audits. These vulnerabilities make IoT networks, which rely on network-level communication for efficiency, prime targets for intrusion attacks, particularly within smart systems and networks (Khanam et al., 2020). IDS play a crucial role in identifying malicious activities within these networks. They provide administrators and users with the means to identify vulnerabilities in computer systems and networks. Intrusion detection methods include anomaly-based detection, which involves simulating and analyzing behavioural patterns to uncover potentially malicious activities. Additionally, hybrid-based detection systems combine signature-based and intrusion-based detection approaches (Alsulami et al., 2022; Islam et al., 2021).

Machine learning-based intrusion detection techniques are essential for bolstering IoT device security. They excel in data monitoring and analysis, efficiently processing large volumes of IoT data. Through algorithmic training, they can detect abnormal patterns, flagging potential threats like irregular network traffic, unauthorized access attempts, and unusual sensor readings. Continuous monitoring is essential for real-time risk identification, facilitating rapid security responses and proactive threat prevention. This approach is critical in today's IoT security landscape to effectively mitigate ever-evolving cyber threats (Khashan et al., 2024).

Moreover, IDS employing behaviour or anomaly detection goes beyond identifying specific attack signatures. Instead, they focus on detecting and analyzing patterns of behaviour that are either malicious or deviate from established norms (Khashan & Khafajah, 2023b). These advanced systems utilize statistical analysis, artificial intelligence, and machine learning techniques to thoroughly examine extensive datasets and network traffic, with the primary goal of flagging any deviations or irregularities (Gamage & Samarabandu, 2020).

IoT devices frequently share networks with other corporate devices, heightening the risk of security breaches. To mitigate this risk, network segmentation is recommended (Khashan et al., 2023a). By segregating IoT devices from other devices, the attack surface can be limited, thereby minimizing the potential consequences of security breaches. Achieving this segregation involves isolating IoT devices from other network components (Sufyan et al., 2023). Machine learning techniques can be employed to monitor network traffic between IoT devices and other devices, rapidly detecting any attempts to breach network segmentation, as illustrated in the study conducted in (Gupta et al., 2020). Additionally, machine learning algorithms can be trained on threat intelligence data to identify and address established security risks, such as common IoT device malware infections. Predictive analysis through machine learning can anticipate security issues by examining historical data and identifying trends before breaches, including insider threats that traditional methods struggle to detect (Albulayhi et al., 2022). On the other hand, in the realm of big data, conventional IDS present challenges, leading to the utilization of machine learning for efficient threat detection and classification in monitoring emerging threats. These challenges are exacerbated by the need for robust technology and refined AI-powered algorithms to process extensive data (Khashan, 2021a; Sarker et al., 2020).

In this study, we aim to enhance the efficacy of intrusion detection methods through the introduction of an advanced heterogeneous machine learning-based stack classifier model. Our primary objective is to elevate the overall performance and accuracy of intrusion detection systems, particularly in identifying and preventing security breaches. Throughout the study, we will thoroughly examine various machine learning algorithms and techniques for intrusion detection to assess their effectiveness. The research also endeavours to explore opportunities for optimizing the efficiency of these methods.

The research contributions are as follows:

1) We conduct a thorough analysis of current intrusion detection methods, identifies limitations, and provides insights for researchers and practitioners in the intrusion detection domain.
2) We present a taxonomy of machine learning techniques for detecting malicious user attacks, focusing on the TON-IOT dataset, contributing valuable insights into IoT intrusion detection.
3) We introduce a heterogeneous machine learning-based stack classifier model for IoT intrusion detection, incorporating feature selection and ensemble modelling to improve key classification metrics. This model achieves exceptional accuracy and effectively addresses IoT network security concerns.

The remainder of this paper is organized as follows: Section II presents background and an in-depth examination of related studies. In Section III, the study's methodology is thoroughly discussed. Section IV discusses the implementation details. Section V reveals the research results and discussions. Section VI concludes the paper and suggests future directions.

2. Background and related work

2.1. ML-Based intrusion detection for IoT security

In the realm of IoT security, researchers have explored innovative approaches to enhance intrusion detection methods. (Kiran et al., 2022) specifically focus on enhancing security in the context of IoT networks. They propose an efficient pairing-free identity-based signcryption scheme using elliptic curve cryptography, showcasing improved confidentiality and authenticity while addressing potential intrusions. Performance analysis and comparisons substantiate the scheme's efficacy in fortifying security within the cloud-assisted IoT environment. Additionally, in (Halimaa & Sundarakantham, 2019), various methods were employed to boost the performance of intrusion detection systems (IDS), primarily focusing on the analysis of extensive network traffic data. The proposed methodology introduces a well-structured classification approach using common machine learning techniques such as Support Vector Machine (SVM) and Naïve Bayes (NB), with SVM displaying superior performance. Comparative analysis is then conducted to evaluate the accuracy and misclassification rates of these methods. In another study in (Alzubaidi et al., 2021), the authors explored the use of ensemble learning techniques, such as random forests (RFs) and gradient boosting for IoT intrusion detection. Their research focused on developing models that improve detection precision while reducing false positives. Nevertheless, these techniques require a significant training data remain vulnerable to adversarial attacks, posing challenges in the development of IoT-based IDS. In a study by (Maniriho et al., 2020), they introduced an anomaly-based approach for IoT networks. Their methodology integrates a hybrid feature selection engine to identify essential features along with the RF algorithm to classify traffic into normal and anomalous categories. To evaluate its performance, they employed the IoTID20 dataset, a recent collection of anomaly detection data in the IoT environment. The study in (Cvitić et al., 2021) developed a classification model for IoT devices in smart homes, achieving a remarkable 99.79% accuracy. By leveraging logistic regression and supervised machine learning, the model uses 13 network traffic features to dynamically categorize devices, providing a robust foundation for monitoring diverse IoT environments.

Training supervised learning algorithms involves using labelled data, which includes instances of typical network activity and identified security threats. Once trained, the system can detect similar threats in real-time. SVM is a widely used supervised learning technique for intrusion detection. SVM achieves its objective by creating a boundary between normal and anomalous network activity. It raises an alarm whenever it detects network traffic that falls outside this boundary (Rani & Kaushal, 2020).

Unsupervised learning algorithms can detect anomalous activity in a network without requiring labelled data or direct supervision. These algorithms are capable of identifying deviations from typical network behaviour and alerting administrators when anomalies occur. The k-means clustering technique is an example of an unsupervised learning algorithm commonly employed in intrusion detection. The k-means algorithm groups similar network traffic patterns together, and it triggers an alarm when it encounters a new instance of network traffic that doesn't fit into any of the established clusters (Gyamfi & Jurcut, 2022). In contrast, deep learning-based techniques for detecting network intrusions leverage neural networks to analyze network traffic patterns. These networks have the capability to grasp complex relationships within the data and identify patterns that may not be immediately apparent to human analysts (Ullah & Mahmoud, 2021). Recurrent Neural Networks (RNNs) represent a prominent intrusion detection approach rooted in deep learning. RNNs are particularly effective in detecting attacks that occur over an extended duration, such as Distributed Denial of Service (DDoS) attacks (Putchala, 2017).

Intrusion detection accuracy can be significantly enhanced through the utilization of ensemble-based approaches, which integrate multiple machine-learning algorithms into a unified detection process (Le Jeune et al., 2021). These algorithms' outcomes are averaged before a final decision is reached. A prevalent method for ensemble-based intrusion detection is the RF algorithm, which aggregates results from various decision trees (DTs). The adoption of ensemble-based intrusion detection approaches has demonstrated a substantial improvement in intrusion detection accuracy compared to using a single algorithm (Khraisat et al., 2020).

Hybrid-based intrusion detection approaches combine several machine learning algorithms with rule-based or expert systems to detect known attack patterns based on predefined criteria. When an attack is identified, the rule-based system triggers an alarm, and a machine learning algorithm conducts further investigation. An example of this approach is the Genetic Algorithm-based Intrusion Detection System (IDS), which employs genetic algorithms to enhance the rule set used by the rule-based system (Awotunde et al., 2021). Furthermore, intrusion detection techniques based on machine learning offer a practical means to enhance the security of the Internet of Things (IoT). Methods such as supervised learning, unsupervised learning, deep learning, ensemble-based detection, and hybrid-based detection can be utilized to

identify and prevent security threats in real-time (Islam et al., 2021). (Bouneb, 2022) contribute to the discussion on hybrid-based intrusion detection approaches. Their research defines and generates groups in IoT systems, allowing processes to be part of multiple groups simultaneously. This innovative approach aligns with the hybrid strategy of combining various machine learning algorithms with rule-based systems for effective intrusion detection. The study not only establishes a theoretical foundation but also demonstrates the practical application of the algorithm through an IoT-based intersection traffic light management system, highlighting its versatility in real-world scenarios. In a study conducted by (Roy et al., 2022), a lightweight intrusion detection framework utilizing machine learning techniques was introduced to accurately detect cyber-attacks and anomalies in resource-constrained IoT networks. Through a series of optimization techniques, including multicollinearity elimination, sampling, and dimensionality reduction, the model efficiently identifies crucial features for intrusion detection, reducing the need for extensive training data and time. Extensive experimentation on the CICIDS2017 and NSL-KDD datasets demonstrated the effectiveness of the proposed approach, with the model achieving a notable level of detection accuracy while maintaining a low false alarm rate.

2.2. Comparative analysis of ML-Based intrusion detection in IoT

According to the study in (Bhuiyan et al., 2021), there is a growing awareness of the security risks associated with the widespread adoption of IoT devices, including issues like unauthorized access, data breaches, and malware attacks. In response to these challenges, machine learning algorithms have risen in importance for intrusion detection within the IoT ecosystem, owing to their ability to analyze extensive datasets and identify security breaches effectively. Among the various machine learning techniques available, artificial neural networks (ANNs) and SVMs have emerged as the favoured options for identifying vulnerabilities in IoT systems. The authors in (Gad et al., 2021) introduced an IDS model tailored for vehicular Ad-hoc networks (VANETs), utilizing the ToN-IoT dataset as the foundational network dataset. This IDS model comprised several essential components, and they conducted training and testing using the same dataset. Their extensive evaluation of various machine learning techniques demonstrated that XGBoost achieved exceptional performance in both binary and multi-class classification scenarios, outperforming all other methods in terms of effectiveness. The researchers in (Taher et al., 2019) employed supervised machine learning techniques to classify network traffic into malicious or normal categories, identifying that a combination of supervised learning algorithms and feature selection methods, particularly utilizing ANN with wrapper feature selection, yielded superior results in intrusion detection success rate compared to the SVM, using the NSL-KDD dataset for evaluation. This study contributes to enhancing intrusion detection accuracy and resource efficiency, offering cost-effective solutions for network security while improving overall network performance. In a study by (Churcher et al., 2021), various machine learning techniques, including KNN, SVM, DT, NB, RF, ANN, and logistic regression (LR), were evaluated for their suitability in IDSs using IoT datasets. Experimental comparisons of these algorithms were conducted, considering parameters such as accuracy, precision, recall, F1 score, and log loss. The results proved that the RF algorithm outperformed others in binary classification across different attack types.

The authors in (Abdullahi et al., 2022) conducted a study addressing challenges posed by the limited computational power of IoT devices, including constrained processing capabilities, limited memory, and finite battery life. These constraints can hinder the execution of advanced machine learning algorithms, particularly in scenarios with limited connectivity, complicating data collection and transmission. The acquisition of labelled data for training IoT-based machine learning models can be expensive and time-consuming, further challenging model development. Another study (Fortuna et al., 2022) delved into smart infrastructure systems that leverage advanced technologies like AI to improve performance and adaptability. The authors emphasized AI's advantages in resource allocation, predictive maintenance, and real-time decision-making for modern infrastructure systems grappling with issues like aging infrastructure and climate change. However, it's crucial to note that the accuracy of machine learning algorithms depends on the quality of data, and challenges such as packet loss, sensor failures, and network congestion can lead to noisy or incomplete IoT data, potentially diminishing the effectiveness of IDS.

The study by Li et al. (2022) proposed a two-stage approach for network-based anomaly detection in smart homes. In the first stage, they clustered devices based on network traffic patterns, while the second stage employed a RF classifier to identify anomalous clusters. Their approach demonstrated superior performance in terms of detection accuracy and efficiency when evaluated using a real smart home dataset. Furthermore, an SVM linear kernel model utilizing UCC data exhibited strength in detecting IoT-based SH HAN attacks. In a separate investigation conducted by (Iftikhar et al., 2022),

an intelligent approach for detecting malicious intrusions in IoT networks was introduced. They utilized a dataset comprising both normal and attack traffic for their analysis. Their findings revealed that the RF with Information Gain and Passive Aggressive Classifier emerged as the top-performing methods in anomaly detection. (Jayalaxmi et al., 2022) conducted a comprehensive survey of machine and deep learning techniques for IoT network intrusion detection and prevention. They emphasized the effectiveness of supervised machine learning, particularly DT and RF algorithms, which achieved an impressive 99% accuracy rate with minimal false positives. However, they also noted that supervised models may experience reduced detection rates for previously unseen attacks. Additionally, the study in (Sharma & Sharma, 2022) delved into security challenges by categorizing and analyzing attacks on resource-constrained IoT devices. Notably, the study brought attention to prevalent threats like social engineering and DoS. Furthermore, the research extensively explored holistic security solutions, encompassing cryptography, blockchain, software-defined networks, and machine learning techniques. (Saheed et al., 2022) presented a machine learning-based intrusion detection system for IoT networks. Their system achieved high accuracy in detecting various attack types but did not address adversarial attacks. They employed PCA for dimensionality reduction and evaluated the model using the modern UNSW-NB15 dataset. In Abbas et al.'s study (Abbas et al., 2021), the evolution of the IoT landscape was explored. To counter escalating cyber threats, this study introduces an ensemble-based intrusion detection model, outperforming existing techniques on the CICIDS2017 dataset with significant improvements in accuracy for both binary and multi-class classification. Furthermore, (Verma et al., 2021) proposed a binary classifier approach using a machine learning ensemble method, specifically the gradient boosting machine (GBM). Trained on pre-processed data packets, this approach effectively detects anomalies and prevents zero-day attacks in IoT networks, showcasing positive class performance metrics with an accuracy of 98.27%, precision of 96.40%, and recall of 95.70%. (Stoian, 2020) explored the applicability of machine learning algorithms in identifying anomalies in IoT network data. Their comparative analysis included RF, NB, and Multi-Layer Perceptron (MLP), with RF yielding optimal results with 99.5% precision. Likewise, (Alsahli et al., 2021) evaluated machine learning algorithms, including RF and K-nearest neighbours (KNN), for identifying attacks on the UNSW-NB15 benchmark dataset. RF and KNN classifiers achieved exceptional accuracy rates of 99%, while NB demonstrated inferior performance. Precision and recall metrics further confirmed RF and KNN's superior performance compared to NB. Table 1 presents a comparison of various machine learning-based intrusion detection schemes.

3. Methodology

3.1. Methodological framework

To fortify the security of the IoT, our research integrates machine learning intrusion detection techniques. Four supervised machine learning classifiers – Naive Bayes (NB), Random Forest (RF), K-nearest neighbours (KNN), and Support Vector Machine (SVM) – are employed alongside a Stack Classifier to analyze the TON-IoT dataset (ToN-IoT). The research workflow commences with meticulous data acquisition from the TON-IoT dataset, followed by preprocessing steps to render it suitable for the classifiers. This preprocessing involves addressing missing values and converting categorical variables into numerical equivalents. The dataset is then partitioned into training and testing sets. The training set becomes the arena for instructing the classifiers - NB, RF, KNN, SVM, and the Stack Classifier. Meanwhile, the testing set serves as the proving ground for evaluating classifier performance. Metrics such as accuracy, precision, recall, and F1-score are employed to scrutinize their efficacy, leading to the identification of the most adept classifier for subsequent analysis, as illustrated in Figure 1.

The incorporation of traditional machine learning models, including NB, RF, KNN, and SVM, is driven by their established efficacy in handling diverse patterns and complexities within IoT data. Renowned for their robustness and ability to capture intricate relationships within data, these models contribute unique strengths to the ensemble model. To address the diverse nature of attacks on IoT networks, a holistic approach is essential. The integration of these traditional models within the ensemble ensures a comprehensive coverage of potential intrusion

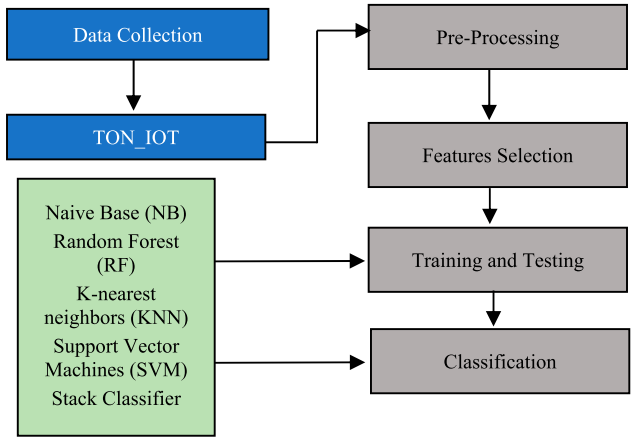

Figure 1. Study Methodological Framework Overview.

Table 1. Comparisons of machine learning-based intrusion detection schemes.

Scheme	Dataset	Key Findings	Limitations
(Abdulla & Jameel, 2023)	NSL-KDD, UNSW-NB15, and CICIDS2017	Provides an overview of different intrusion detection systems and attacks (Probe, R2L, U2R, and DoS) and analyzes the performance of different algorithms such as DT, RF, and boosting algorithms.	Does not use modern datasets, such as Ton-IoT IoTID20, which include features of IoT network traffic.
(Ding et al., 2023)	TON-IoT, Edge-IIoTset, and UNSW-NB15	This paper presents an alternative deep learning model known as DeepAK-IoT to detect cyberattacks against IoT devices. DeepAK-IoT model provided 90.57% accuracy for TON IoT, 94.96% for Edge-IIoTset, and 98.41% for UNSW NB15.	The DeepAK-IoT model has less accuracy on TON-IoT dataset.
(Roy et al., 2022)	CICIDS2017 and NSL-KDD datasets	Identifying cyber-attacks and anomalies in IoT networks using B-Stacking algorithm ensemble learning methods, the model has a high detection rate and a low false alarm rate. B-Stacking accuracy on NSL-KDD dataset is %98.5 and on CICID2017 achieved %99.11.	No compression on benchmark dataset such as ToN-IoT and it is not evaluated on real IoT network. Also, these datasets are not updated with new attacks.
(Gad et al., 2021)	ToN-IoT dataset	The Chi2 technique was used to pick features in a network dataset. Boosting algorithm such as XGBoost perform better than traditional intrusion detection methods in terms of accuracy and false positive rate.	The evaluation was carried out only on a singular dataset, and only two machine learning algorithms were examined. The problem with a class imbalance in the ToN-IoT dataset, as well as missing values.
(Saheed et al., 2022)	UNSW-NB15	Achieves a high level of accuracy in detecting different types of attacks in IoT networks using PCA.	Does not account for adversarial attacks and the author does not take into consideration other cyber threats.
(Jayalaxmi et al., 2022)	Different benchmark datasets	Indicated that the supervised machine learning methodologies show excellent precision in comparison to the unsupervised models. The DT and RF algorithms have demonstrated excellent performance, giving a 99% accuracy rate and minimal false positives	The existing detection models lack in identifying zero-day attacks and result in high false alarms and create impact on the response time of the model.
(Iftikhar et al., 2022)	UNSW-NB15	The study shows that for the proposed method 'RF (Information Gain)' and 'Passive Aggressive Classifier' provide the best cross-validation accuracies.	Methodologies used are only supervised machine learning algorithms.
(Li et al., 2022)	Real smart home environment dataset	Results show with taking into consideration the detection performance and resource consumption, SVM with a linear kernel has the best performance among all models.	XGBoost takes the longest time to generate a detection model and does not cover various types of attacks. It should be tested in complex environments and with different smart home devices
(Churcher et al., 2021)	Bot-IoT	KNN, SVM, DT, NB, RF, ANN, and LR. The accuracy of multi-classification, ANN predicts the type of attack with higher accuracy.	Not able to investigate several threats such as fault injection and jamming to validate the multi-class functionality of models.
(Qureshi et al., 2020)	–	Discussed security issues and their significance on the security and confidentiality of the data.	Lack of availability of sufficient security-related data in IoT networks.
(Alsahli et al., 2021)	UNSW-NB15	Classifications used are RF, KNN, and NB. The study results shows that RF and KNN classifiers give the best performance with an accuracy of 100% (without noise injection) and 99% (with 10% noise filtering), while the NB classifier gives the worst performance with an accuracy of 95.35%.	The work can be expanded in future by studying the performance of other machine learning classifiers using different evaluation metrics on different datasets.
(Stoian, 2020)	IoT-23	The algorithms used are RF, NB, and MLP for anomaly detection in IoT environments.	The main drawback of this study is the presence of technical limitations, specifically the requirement to partition and encoding of the data, which may have impacted the outcomes.
Taher et al. (2019)	NSL-KDD	Researchers used supervised machine learning techniques ANN based Wrapper feature and it performs better than other existing machine learning algorithms such as SVM.	The study is limited by the only use of the NSL-KDD dataset, which may not accurately represent all types of network traffic and potential attacks.
(Hasan et al., 2019)	DS2OS traffic traces database	Machine Learning Algorithms used are RF, SVM, DT, and ANN. The RF algorithm show greater performance compared to other machine learning techniques in terms of accuracy.	No analysis has been carried out to determine the computational resources.
(Abbas et al., 2021)	CICIDS2017	The model, integrating logistic regression, naive Bayes, and decision tree with a voting classifier. The evaluation on the CICIDS2017 dataset illustrates significant enhancements, particularly in both binary and multi-class scenarios.	The limitation lies in the study's focus on the CICIDS2017 dataset, calling for additional assessments across diverse datasets to ensure the model's generalizability to various intrusion scenarios.
(Verma et al., 2021)	CSE-CIC-IDS2018-V2	The binary classifier employed in the study, utilizing a machine learning ensemble method, achieves an accuracy of 98.27%, precision of 96.40%, and recall of 95.70%. This classifier effectively filters malicious traffic in IoT networks, offering robust defense against zero-day attacks.	The study relies on a single dataset for training, validating, and testing the proposed intrusion detection model. Therefore, the findings and the model's performance may vary when applied to different datasets or scenarios not represented in the chosen dataset.

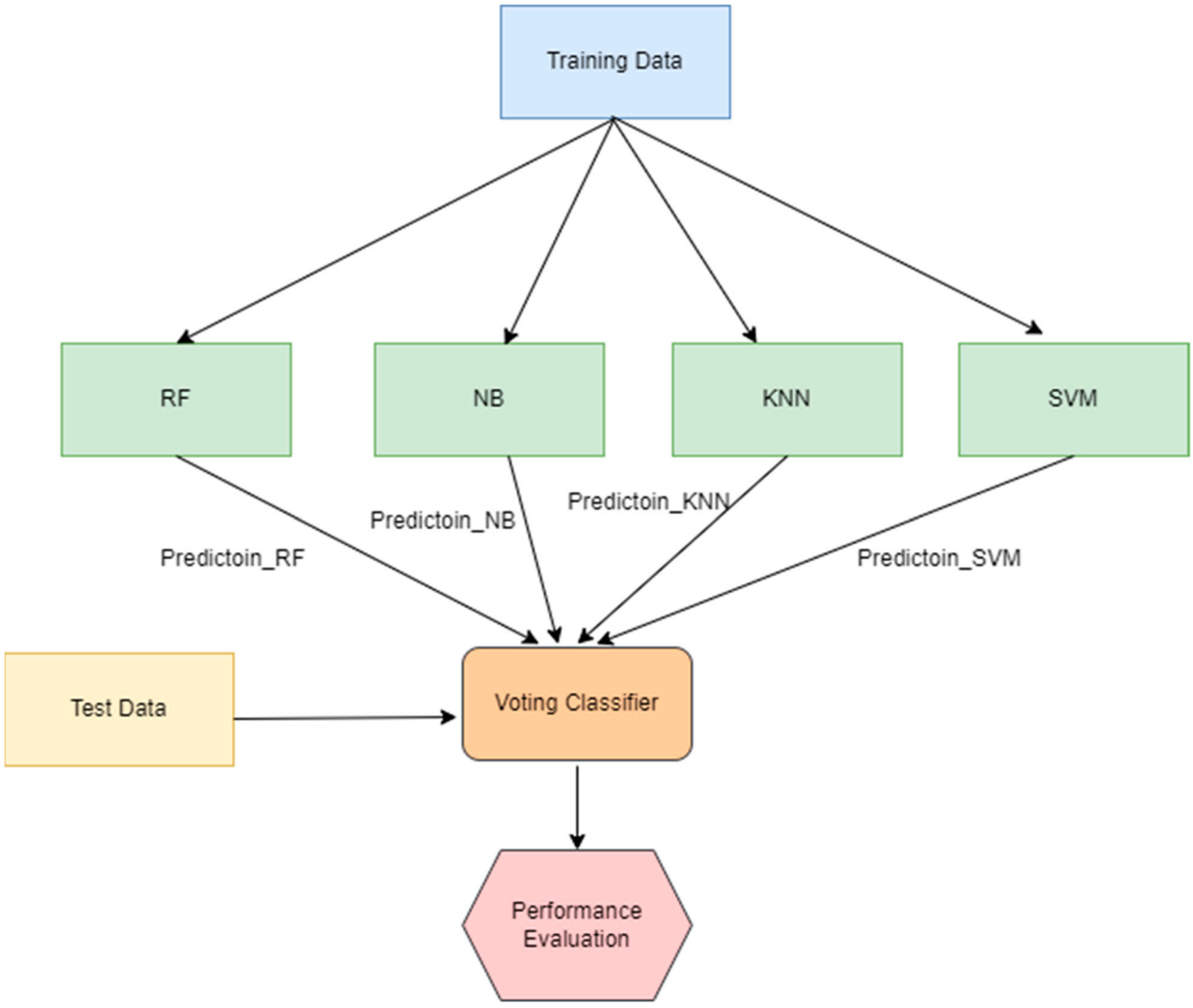

Figure 2. Integrating ML Models and Ensemble Learning for Enhanced Intrusion Detection.

scenarios. Figure 2 illustrates the integration between the employed ML models and Ensemble Learning, enhancing the clarity of the ensemble approach for improved intrusion detection.

Once the most superior classifier is identified through rigorous evaluation, it becomes a pivotal component of our ensemble model. This carefully chosen model, along with the Stack Classifier, forms a synergistic ensemble that leverages the individual strengths of each model. The ensemble approach enhances the overall intrusion detection capability by combining the distinctive abilities of traditional models with the adaptability and learning capabilities of the Stack Classifier. This hybrid model, informed by the collective intelligence of diverse classifiers, provides a robust and versatile defense against evolving threats in the IoT landscape. The subsequent evaluation then benchmarks the effectiveness of this machine learning intrusion detection technique against traditional intrusion detection methods, including rule-based and signature-based approaches, providing a comprehensive understanding of its efficacy.

The selected classifier is then deployed on the TON-IoT dataset for intrusion attack detection. Predictions regarding attack types are made, and these results are rigorously compared with the actual attack types. The classifier's performance is meticulously evaluated based on the percentage of correctly classified instances. Significantly, the research goes beyond the immediate evaluation and juxtaposes the machine learning intrusion detection technique against traditional intrusion detection methods, encompassing rule-based and signature-based approaches. In essence, this research undertakes a comprehensive supervised learning approach to elevate IoT security through machine learning intrusion detection techniques. Beginning with the preprocessing

and partitioning of the TON-IoT dataset, the study progresses to training and assessing the performance of four supervised machine learning classifiers. The superior classifier, determined through rigorous evaluation, is then employed to detect intrusion attacks. The effectiveness of this machine learning intrusion detection technique is subsequently benchmarked against traditional intrusion detection methods.

3.2. Data collection

For this study, we meticulously selected the ToN-IoT dataset due to its exceptional qualities that align with the requirements of our research. The dataset was chosen for its robust representation of real-world IoT scenarios, offering a diverse array of attack types, high data quality, and public accessibility. This comprehensive collection encapsulates both normal and malicious IoT network events, mirroring the multitude of attacks encountered in real IoT environments. The ToN-IoT dataset accurately emulates IoT settings, including a wide spectrum of attack types such as DoS, DDoS, information theft, and reconnaissance. This diversity is instrumental in enabling machine learning algorithms to adeptly identify and categorize various intrusions, learning from a broad spectrum of attack patterns. To harness the full potential of this dataset in conjunction with machine learning algorithms, we embarked on a thorough cleaning and preprocessing phase. During the preprocessing phase, we addressed inconsistencies, removed duplicate entries, handled missing values, and normalized feature values to ensure the dataset's integrity and suitability for machine learning algorithms. This meticulous process enhances the dataset's quality and ensures that our subsequent analyses are built on a solid foundation.

Our choice of the ToN-IoT dataset was deliberate, considering the limitations of alternative datasets. Many other available datasets lack IoT-specific data, contain outdated attack patterns, or offer a limited variety of attack types. The ToN-IoT dataset's comprehensive representation, diversity, and quality make it an ideal candidate for the development and testing of machine learning intrusion detection methods tailored to IoT security. The inclusion of rigorous data cleaning and preprocessing steps further strengthens the dataset's suitability for implementation with machine learning algorithms, ensuring that our study's findings are both robust and applicable to real-world IoT security challenges.

3.3. ML classifiers selection and overview

We have used four supervised machine learning classifiers, namely SVM, RF, KNN, NB, and a Stack Classifier. These classifiers were implemented using scikit-learn, a widely-used Python library for machine learning tasks. Scikit-learn provides a rich set of tools and algorithms for various machine learning applications, making it a versatile choice for our study. The best-performing algorithm for intrusion detection in IoT environments was determined by evaluating the performance metrics of each machine learning algorithm. The following subsections provide an overview of the algorithms that will be applied in this research.

3.3.1. Support Vector Machines (SVM)

The Support Vector Machines aim to find an optimal decision boundary or hyperplane that effectively separates different classes within a given dataset. This is achieved by maximizing the margin, which represents the distance between the hyperplane and the nearest data points from each class.

3.3.2. Random Forest (RF)

Random Forest is an ensemble learning technique that builds multiple decision trees during training and produces the mode of class predictions for classification problems. It is known for its strong performance in intrusion detection and its resistance to overfitting, making it suitable for handling large datasets with numerous features.

3.3.3. Naïve Bayes (NB)

The Naïve Bayes algorithm is a probabilistic classification method based on the Bayes theorem and assumes feature independence. It demonstrates high performance in various categorization tasks, including intrusion detection, and offers a rapid and user-friendly implementation.

3.3.4. K-nearest neighbours (KNN)

The K-nearest neighbours (KNN) algorithm is a non-parametric supervised learning approach commonly used for classification and regression. It predicts the class of a new data point based on the majority class among its K-nearest neighbours in the feature space. Performance optimization can be achieved through hyperparameter tuning.

3.3.5. Heterogeneous Stack Classifier

Stack classifier is a heterogeneous ensemble learning technique that combines multiple classification models to enhance overall performance and predictions. In our approach, we combine the four machine learning models (SVM, RF, NB, and KNN) into an ensemble learning model. The results are determined using a majority voting technique, allowing the combined model to benefit from the strengths of each individual model.

As a conclusion, delving into the integrated machine learning models and the rationale behind their selection,

we carefully chose four supervised classifiers – SVM, RF, NB, and KNN. These classifiers were selected for their proven efficacy in navigating intricate patterns within IoT data, emphasizing robustness by integrating traditional models known for reliability in intrusion detection scenarios. Additionally, a Heterogeneous Stack Classifier amalgamates these individual models, employing a majority voting technique to enhance overall intrusion detection capability. This ensemble strategy ensures a comprehensive defense against a diverse array of attacks on IoT networks. Each model, whether SVM, RF, NB, or KNN, contributes distinct strengths to the collective intelligence of the ensemble, forming a versatile and robust machine learning intrusion detection technique tailored to the challenges of IoT security.

4. Implementation

4.1. Dataset selection

The selection process for the TON_IoT dataset involved consideration of several factors to ensure its utility and representativeness. This dataset encompasses both normal and anomalous traffic samples, with anomalies stemming from various sources, including malicious attacks, device malfunctions, and unusual user behaviour. By incorporating a diverse range of anomalies, the dataset offers researchers a comprehensive array of scenarios for the development and evaluation of anomaly detection methods.

The dataset provides labelled instances of both normal and anomalous traffic, serving as ground truth data for assessing the performance of anomaly detection techniques. This precise labelling facilitates comparisons and benchmarks across different approaches, enabling the measurement of their effectiveness in identifying anomalies. In total, the dataset comprises 46,000 rows and encompasses 45 distinct features. These features include chronological context, IP addresses, ports, communication protocols, and service types, as illustrated in Table 2. Moreover, the dataset includes DNS, SSL, and HTTP features, offering deeper insights into network dynamics and potential vulnerabilities. The incorporation of these features, coupled with ground truth labels, streamlines supervised machine learning tasks, facilitating model training and validation based on network interaction classification.

4.2. Descriptive statistics

In our analysis of the dataset, we conducted a comprehensive examination of descriptive statistics to gain insights into the characteristics of the data. These statistics offer a high-level overview of various attributes within the dataset, shedding light on their distributions and ranges. The dataset encompasses a wide temporal range, reflecting its diverse historical context. Additionally, attributes related to network ports and communication durations exhibit considerable variability, highlighting the dataset's rich diversity. The data also features extensive transmission of bytes and packets, with occasional instances of exceptionally high data transfer.

Furthermore, attributes associated with DNS and HTTP operations demonstrate diverse patterns, providing valuable context for understanding network behaviour. The label distribution reveals that approximately 34.93% of network interactions are classified as anomalies, underscoring the importance of intrusion detection in this dataset. In addition, our examination of object-type columns provided key information about source and destination IPs, network protocols, services, connection states, and potential network anomalies, contributing to a holistic understanding of the data.

Table 2. Features Description Overview.

Attribute	Description
ts	Provides chronological context to the data, serving as a resource for studying temporal patterns and devising time-based models of IoT behaviour.
src_ip, dst_ip	Source and destination IP addresses which constitute the basis of the network interaction.
src_port, dst_port	Source and destination ports which, along with IP addresses, give an intricate mapping of the communication architecture.
proto	Communication protocol used in the network interaction.
service	Indicates the type of service being availed by a network interaction, helping investigate service-oriented vulnerabilities.
duration, src_bytes, dst_bytes, conn_state, missed_bytes, src_pkts, src_ip_bytes, dst_pkts, dst_ip_bytes	Features giving a detailed panorama of data transmission and reception characteristics, including connection state and packet details, allowing for deep inspection of data exchange health and potential anomaly detection.
dns_, ssl_	Features catered to DNS and SSL services. These provide a refined analysis of the security mechanisms deployed in the IoT network, highlighting potential points of exploitation.
http_∗	Features that extend to HTTP protocol dynamics, allowing researchers to extrapolate the behaviour of the IoT network in web communication scenarios and pinpoint possible HTTP-related vulnerabilities.
weird_∗	Attributes capturing various irregularities in the network communication, aiding in the identification of abnormal behaviours and security threats.
label, type	The ground truth for any supervised machine learning task. They provide the classification of the network interaction, making them pivotal for model training and validation.

4.3. Data visualization and categorical value conversion

We utilized the Plotly.express library (Plotly Library) to generate a pie chart that visually represents the distribution of attack types within the dataset. By analyzing the counts of each unique value in the 'label' column of the dataframe (denoted as df), this visualization provides valuable insights into the occurrence of various attack and normal types. The pie chart offers a clear representation of the relative frequencies or percentages of both attack and normal types, with each slice's size corresponding to the proportionate count of the respective label category ('0′ for normal and '1′ for attack). This visual insight aids in understanding the dataset's composition.

Additionally, within the context of data preprocessing, we address the need to transform categorical columns into numerical values. This crucial transformation is achieved through the use of Label Encoder, a technique that assigns unique numerical codes to categorical labels. Label Encoder ensures that categorical information is accurately represented for compatibility with machine learning algorithms that require numerical inputs. The ToN-IoT dataset contains multiple categorical features, all of which undergo conversion into numerical representations using Label Encoder. This step is essential for preparing the data for subsequent machine learning analysis, contributing to the overall data readiness for intrusion detection.

4.4. Identifying and handling highly correlated features

We employ Pearson correlation, a widely-used statistical measure, to assess the linear relationships between features in our dataset. The Pearson correlation coefficient, ranging from -1 to 1, allows us to quantify the strength and nature of these relationships:

- A correlation coefficient close to 0 indicates a weak or no connection between variables.
- A coefficient approaching 1 signifies a strong positive correlation, indicating that variables move in the same direction.
- A coefficient nearing -1 indicates a strong negative correlation, suggesting that variables move in opposite directions.

To visualize these correlations, we create a Pearson correlation matrix for our dataframe and use a heatmap plot. The heatmap visually highlights highly correlated features, with darker colours indicating stronger positive or negative correlations, while lighter colours denote weaker or no correlation. The diagonal line in the plot represents each feature's correlation with itself, always yielding a value of 1.

Identifying and handling highly correlated variables is crucial as they can introduce redundancy within the dataset, potentially hindering model performance. This process plays a pivotal role in mitigating multicollinearity, where predictor variables exhibit strong linear relationships with each other. Reducing multicollinearity enhances model reliability and robustness, facilitating better interpretation of predictor-target relationships. This step significantly improves the accuracy and stability of machine learning algorithms when dealing with highly correlated features.

Additionally, we employ MinMaxScaler to normalize the features. MinMaxScaler scales the features to a specified range, typically between 0 and 1. Normalization ensures that all features have a similar scale and prevents any feature from dominating the learning algorithm based solely on its magnitude. It brings the values of different features to a common scale, allowing them to contribute equally during the learning process. This combined approach, as shown in Figure 3, simplifies the feature space, ensures the model is trained with informative, non-redundant features, and ultimately enhances predictive accuracy and model performance.

4.5. K-Best feature selection technique

The K-Best feature selection technique plays a crucial role in optimizing our model's performance by systematically identifying the top *k* most relevant features from the given dataset. This method strategically reduces data dimensionality, enhancing interpretability, and mitigating overfitting. Employing the SelectKBest method, features are rigorously assessed through the f_classif scoring function, an implementation of the ANOVA F-value between label/outputs and features/inputs. A SelectKBest object is instantiated with f_classif as the scoring function, selecting the 15 most critical features, as indicated in Table 3 and Figure 4. The subsequent application of the 'fit_transform' method ensures the feature selector is well-fitted to the data, retaining only the top features based on their ANOVA F-value scores. This meticulous process significantly improves the model's predictive accuracy.

The criteria for selecting the top 15 features with the K-Best algorithm are grounded in their profound impact on IoT security. These features are considered more critical due to their exceptional ability to effectively distinguish between normal and malicious activities within IoT networks. The criteria prioritize features

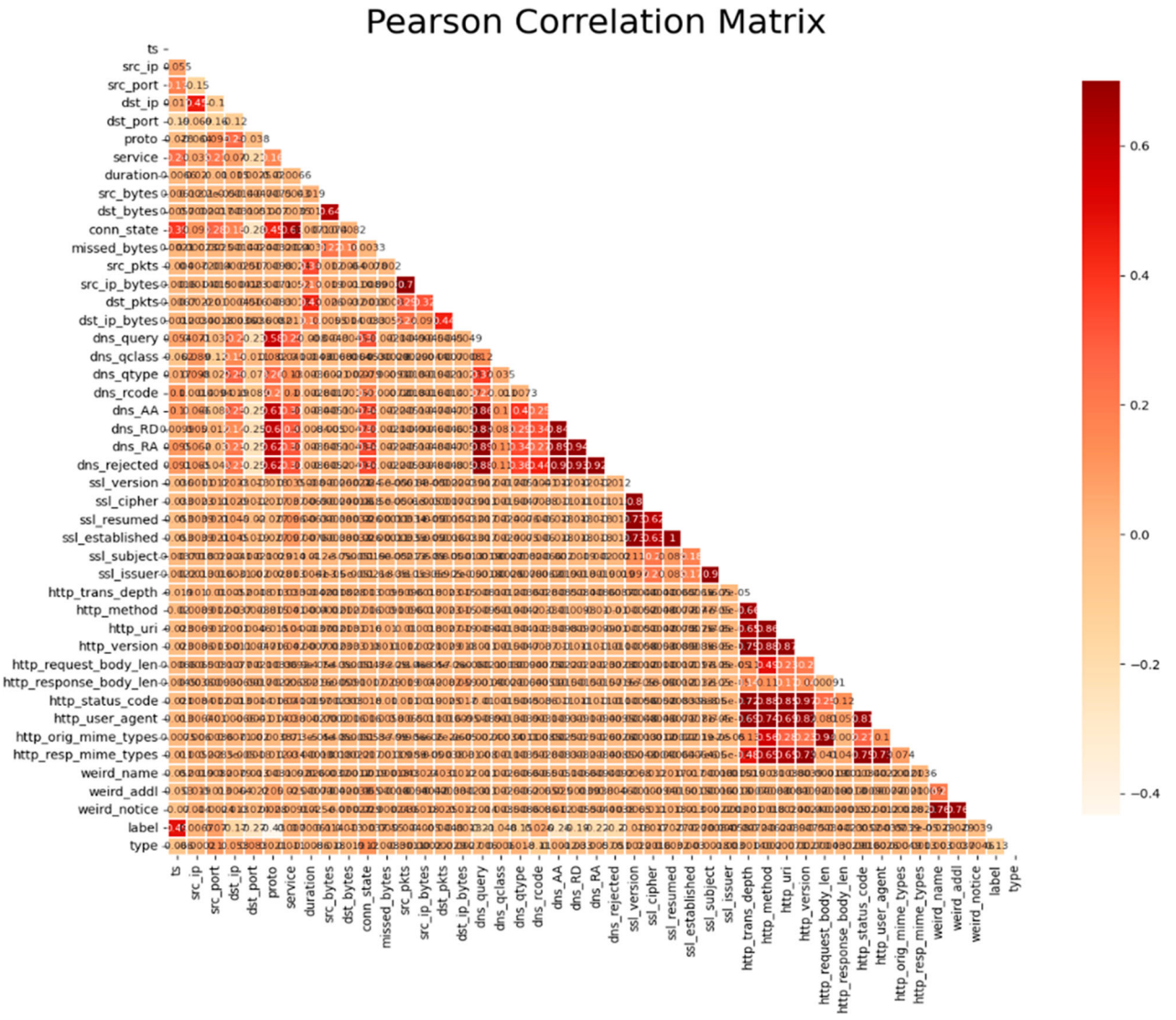

Figure 3. Highly Correlated Feature Selection in Pearson Correlation Graph.

Table 3. Overview of Selected Features Using SelectKBest.

Feature	Score
ts	144747.649
src_port	2240.739934
dst_ip	14308.91563
dst_port	36482.15187
proto	106413.2546
dns_query	21480.6171
dns_qclass	1064.45858
dns_qtype	9906.244769
dns_AA	33690.10914
dns_RD	17740.62108
ssl_resumed	337.1998768
weird_name	380.5708612
weird_addl	399.5059895
weird_notice	688.3944596
type	7459.265943

with high significance in the context of intrusion detection. The K-Best algorithm serves as a precise tool to identify these crucial features, contributing to a more robust defense against evolving threats. This transparent and meticulous feature selection process ensures that the model exclusively focuses on the most pertinent aspects of IoT data, reinforcing the overall efficiency of our security framework. In the context of model optimization, the integration of the K-Best feature selection technique assumes paramount importance. This systematic evaluation ensures the model focuses on the most relevant features from IoT data, particularly vital for intrusion detection. Additionally, the seamless integration of ensemble modelling harnesses the collective intelligence of various machine learning models, offering a robust defense against diverse intrusion scenarios. The combined use of K-Best feature selection and ensemble modelling proves instrumental in fortifying IoT security, providing an advanced and adaptive defense mechanism against evolving threats. This comprehensive approach enhances the model's ability to detect and respond to

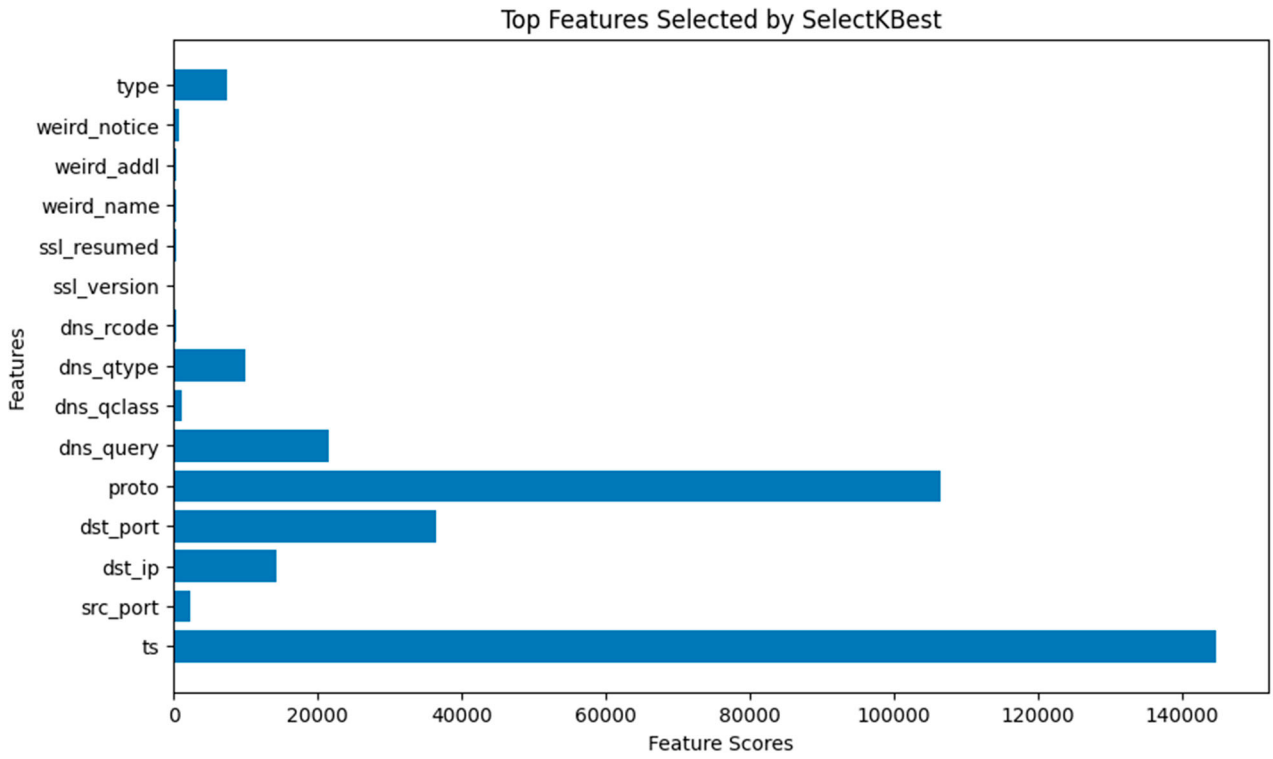

Figure 4. Implementation of K-Best Feature Selection Technique.

emerging security challenges in the dynamic landscape of IoT.

5. Results and discussions

In the comprehensive evaluation of network intrusion detection using machine learning models, we employ crucial performance metrics, namely accuracy, precision, recall, and the F1-score. Accuracy stands as a measure of the model's ability to predict real-world outcomes accurately. Precision assesses the reliability of negative predictions, emphasizing the importance of minimizing false positives, which are particularly critical in the context of intrusion detection. Recall, or the true positive rate, measures the model's effectiveness in identifying positive examples, specifically actual network intrusions. Achieving high recall values is imperative to minimize the risk of missed detections or false negatives, ensuring a robust intrusion detection system.

The F1-score, a harmonic mean of precision and recall, provides a consolidated assessment of overall model performance. A higher F1-score indicates superior performance, considering both accuracy and completeness. These carefully selected metrics collectively contribute to the thorough evaluation of the effectiveness of our machine learning-based network intrusion detection systems. We focus on precision and recall to ensure a balanced approach, emphasizing the model's capability to accurately identify and mitigate potential threats.

Table 4. Technical Details of Utilized Machine Learning Models.

Model	Hyperparameters	Preprocessing Steps
RF Classifier	Number of Trees: 100, Maximum Depth: 10, Minimum Samples per Leaf: 1	Standardization, Handling Missing Values
NB Classifier	Default settings (No hyperparameter tuning)	Feature Scaling (Min-Max Scaling), Handling Missing Values (Simple Imputation)
SVM Classifier	Kernel: RBF, Regularization: 1.0	Feature Scaling
K-Nearest Neighbours	Number of Neighbours: 5, Weight Function: Distance	Feature Scaling
Stack Classifier	Ensemble of RF, NB, SVM, and KNN	Individual Model Tuning, Feature Scaling

Table 4 presents the technical specifications of the machine learning models employed in this study. Our evaluation entails a detailed examination of these metrics, offering a nuanced insight into the performance of the model. Through careful consideration of these factors, our goal is to furnish a comprehensive and transparent analysis of the ensemble model's effectiveness in the intricate domain of network intrusion detection.

5.1. RF classifier results

The Random Forest (RF) classifier demonstrated outstanding performance in its application to the ToN-IoT dataset for intrusion detection. It achieved an impressive accuracy rate of 98.20%, underlining its ability to

effectively differentiate between benign and malicious network traffic, a pivotal aspect of IoT network security. This high level of accuracy substantially reduces the risk of missing potential threats or generating false alarms, ensuring a more secure IoT environment. Additionally, the RF model exhibited a precision rate of 98.19%, signifying its precision in diagnosing intrusion events and minimizing the likelihood of misclassifying normal activities as intrusions. This precision enhances the model's reliability and its potential to identify true intrusion instances. Furthermore, the RF classifier's recall rate of 98.19% emphasizes its proficiency in recognizing genuine intrusion events within the dataset, further fortifying its security capabilities. Moreover, the model's 98.18% F1-score, a balanced metric that combines precision and recall, illustrates its overall effectiveness in intrusion detection. In summary, the Random Forest classifier's exceptional performance across these key metrics underscores its robustness in IoT network intrusion detection, making it a valuable asset for enhancing security in IoT environments.

5.2. NB classification results

The NB classifier's performance in network intrusion detection using the ToN-IoT dataset is both noteworthy and informative. The NB model achieved an accuracy of 80.46%, signifying its capability to effectively categorize network traffic instances as normal or invasive. This accuracy is pivotal in ensuring that potential threats are identified while minimizing false alarms, thus contributing to a more secure IoT environment. The model further demonstrated a precision rate of 87.45%, indicating its accuracy in identifying intrusion events and reducing the likelihood of classifying normal activities as intrusions. The model's impressive recall rate of 80.45% emphasizes its proficiency in recognizing genuine intrusion instances within the dataset, reinforcing its security capabilities. Additionally, the NB model's F1 score, a balanced metric that combines precision and recall, reached 81.44%, showcasing an overall effective performance.

The confusion matrix in Figure 5 provides insights into the model's performance, where true positives amounted to 52,387 instances, correctly identifying network invasions. False positives were 22,516, representing cases where normal network activity was misclassified as intrusion. Notably, there were no false negatives, indicating that the model correctly identified all legitimate network traffic while disregarding intrusion attempts. These findings collectively underscore the NB classifier's competence in network intrusion detection, positioning it as a valuable asset in enhancing IoT network security.

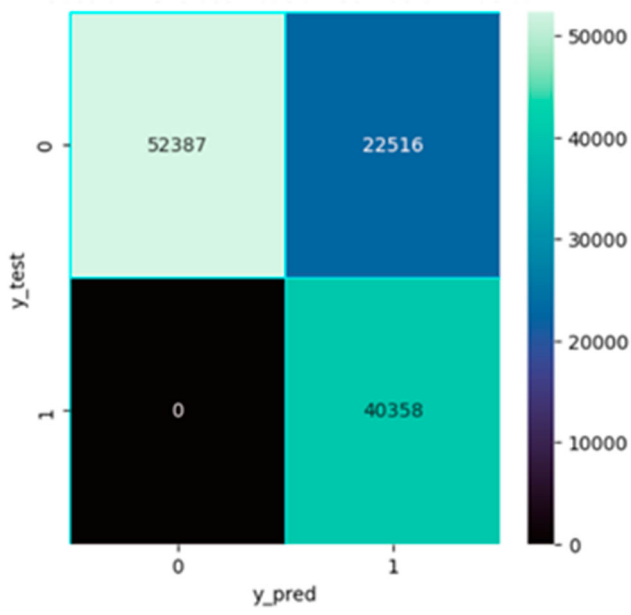

Figure 5. Naïve Bayes Confusion Matrix.

5.3. SVM classification results

The results of the SVM classifier applied to the ToN-IoT dataset for intrusion detection reveal that the SVM model demonstrated an impressive accuracy of 98.10%, highlighting its ability to effectively distinguish between benign and malicious network traffic. This high accuracy rate is critical for ensuring the security of IoT networks, as it minimizes the chances of overlooking potential threats or generating false alarms. Additionally, the SVM model exhibited a precision rate of 95.10%, emphasizing its accuracy in correctly identifying intrusion events and reducing the likelihood of classifying normal activities as intrusions. The SVM model's recall rate also reached 98.10%, indicating its proficiency in recognizing genuine intrusion instances within the dataset and enhancing its security capabilities. Furthermore, the SVM model achieved an F1-score of 98.09%, which is a balanced metric encompassing both precision and recall, further illustrating its overall effective performance in intrusion detection. These results collectively underscore the competence of the SVM classifier in enhancing IoT network security through reliable intrusion detection.

5.4. K-Nearest neighbors results

The KNN classifier exhibits exceptional performance when applied to the ToN-IoT dataset for intrusion detection. With an impressive accuracy of 99.90%, the KNN model excels in accurately categorizing network traffic instances as normal or invasive, minimizing the risk of misclassification and enhancing IoT network security. The

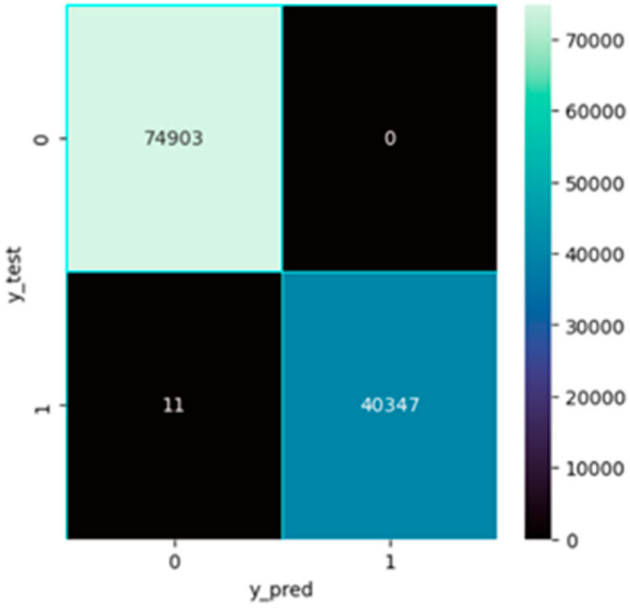

Figure 6. K-nearest Confusion Matrix.

model's precision rate of 99.90% further emphasizes its precision in classifying intrusion events, indicating rare occurrences of false positives. Notably, the KNN model achieves a remarkable recall rate of 99.91%, signifying its proficiency in recognizing genuine intrusion instances within the dataset while minimizing false negatives. The high F1-score of 99.90% underscores the model's balanced performance in terms of precision and recall.

The accompanying confusion matrix, as depicted in Figure 6, reveals 74,903 true positives, correctly identifying network invasions, and zero false positives for benign network activity, reflecting the model's robustness. While there were 11 false negatives, instances of benign activity misclassified as intrusions, the KNN classifier's overall performance remains exceptional.

5.5. Proposed Stack Classifier evaluation and comparisons

In this evaluation, we delve into the outcomes of our meticulously trained heterogeneous Stack Classifier model on the ToN-IoT dataset, prominently illustrated in Figure 7. Impressively, the proposed Stack Classifier achieved an outstanding accuracy rate of 99.99%, demonstrating its exceptional prowess in accurately categorizing network traffic instances as normal or intrusive. This remarkable accuracy was complemented by a precision of 99.98%, indicating the model's precision in minimizing false positives while maintaining high accuracy in intrusion detection.

Furthermore, the model showcased a recall rate of 99.99%, signifying its effectiveness in correctly identifying true intrusions while keeping false negatives at an astonishingly low level. The F1-score, which balances precision and recall, reached an outstanding 99.98%, underscoring the Stack Classifier's well-rounded performance.

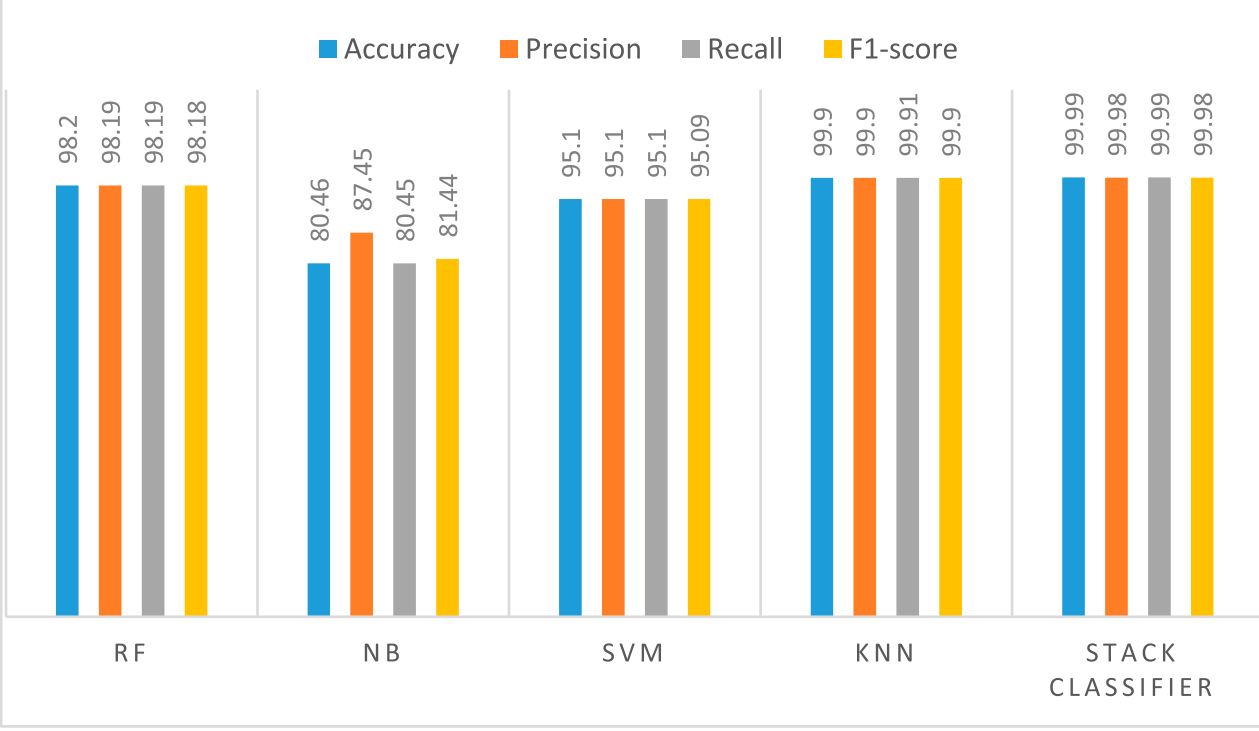

Figure 7. Comparative analysis with the proposed Stack Classifier model.

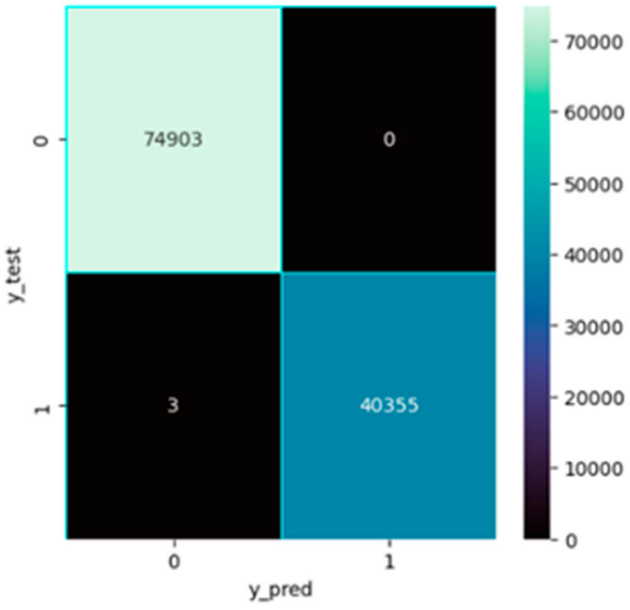

Figure 8. Confusion matrix of proposed Stack Classifier model.

To gain a comprehensive understanding of the results, we begin by examining the performance of our Stack Classifier, as illustrated in the confusion matrix in Figure 8. The Stack Classifier showcased remarkable performance, correctly identifying 40,355 intrusion occurrences (TP) and 74,903 normal instances (TN). This translated to a flawless detection rate with zero false positives. However, it is important to note that the model missed three genuine intrusions, resulting in three false negatives.

Moving beyond individual model assessments, our discussion extends to the practical implications of implementing intrusion detection systems in IoT networks. The seamless integration of the Stack Classifier into existing IoT infrastructures becomes a pivotal consideration, given its adaptability and robustness for real-world applications. Integrating the proposed model involves meticulous considerations of existing network architectures to ensure harmonious assimilation. This entails identifying key integration points, addressing compatibility with diverse IoT devices, and establishing effective communication protocols.

Our focus then shifts to a comparative analysis between our proposed Stack Classifier model and the 'DeepAK-IoT' model presented by (Ding et al., 2023). Illustrated in Figure 9, the visual representation showcases accuracy comparisons. Ding et al. reported an accuracy rate of 90.57% for their 'DeepAK-IoT' model when applied to the TON IoT dataset. In contrast, our ensemble approach, leveraging a combination of models, demonstrated accuracy scores ranging from 80.46% to an impressive 99.99%. Notably, our Stack Classifier model outperformed all others, achieving the highest accuracy at 99.99%. This comprehensive analysis not only underscores the technical superiority of our model but also emphasizes its practical viability for seamless integration into diverse IoT environments, further solidifying its potential for real-world application.

In summary, our comparative analysis underscores the Stack Classifier's consistent superiority over individual models, such as KNN, RF, SVM, and NB, across pivotal metrics like accuracy, precision, recall, and F1-score. While KNN closely follows the Stack Classifier, RF and SVM exhibit robust results, and NB lags with comparatively lower accuracy. Recognizing the paramount importance of accuracy, we also acknowledge the significance

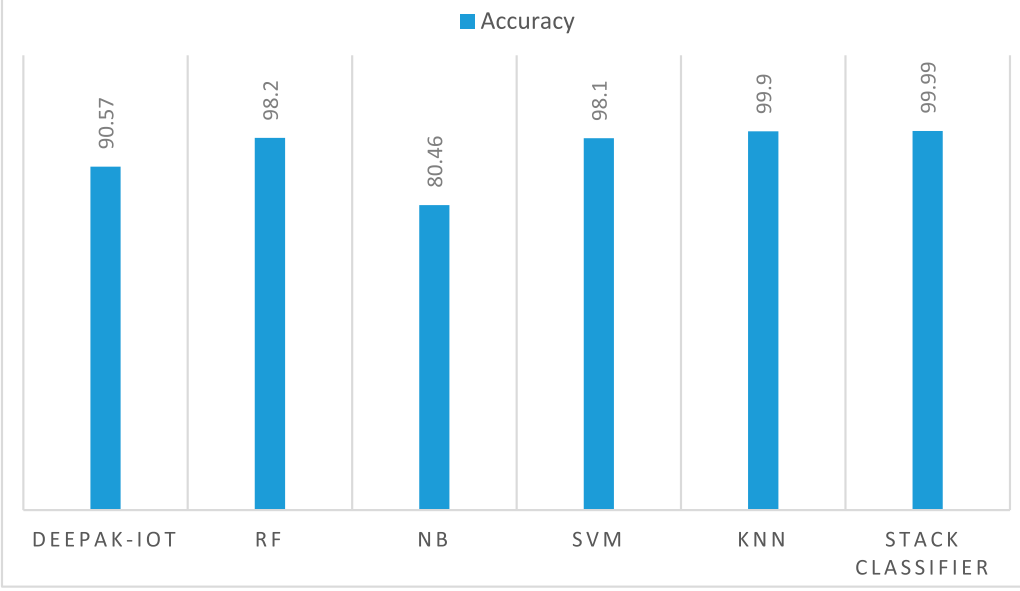

Figure 9. Comparative analysis with related models.

of considering various metrics and contextual factors in evaluating model efficacy. This comprehensive analysis serves as a valuable reference for researchers, practitioners, and decision-makers, providing nuanced insights for informed decision-making based on specific requirements, performance goals, and trade-off preferences.

Furthermore, beyond its technical prowess, the Stack Classifier's practical adaptability enhances its appeal for real-world implementation. As our analysis unfolds, we establish not only its consistent superiority over individual models but also shed light on its broader implications. The nuanced performance assessment of the ensemble approach, excelling not only in accuracy but also in maintaining a balance across precision, recall, and F1-score, underscores its robustness in minimizing both false positives and false negatives. This dual emphasis on technical excellence and practical effectiveness positions the Stack Classifier as an invaluable asset for intrusion detection in the intricate landscape of IoT environments. The comprehensive evaluation extends beyond technical metrics, offering a holistic and effective solution for IoT network security. Its consistent high performance across precision, recall, and F1-score, coupled with exceptional accuracy, solidifies the Stack Classifier's standing as a reliable and practical choice for researchers, practitioners, and decision-makers in their pursuit of enhanced security in complex IoT environments. The detailed insights provided herein serve as a valuable guide for selecting a model that aligns seamlessly with specific security requirements and performance goals.

6. Conclusion and future work

This study has effectively addressed the critical challenge of intrusion detection within IoT networks by introducing a powerful Stack Classifier model. Our model, thoughtfully designed to incorporate the K-Best feature selection algorithm and ensemble modelling, has been instrumental in optimizing essential classification metrics for detecting intrusions in IoT data. Through the strategic selection of the top 15 critical features and the adept integration of various traditional machine learning models into an ensemble, our approach has been able to capture valuable information from these features and harness the strengths of individual models. The results of our extensive experimental evaluations, conducted using the 'Ton IoT dataset' and comparing our ensemble model with individual models, consistently demonstrated exceptional performance. The ensemble model consistently outperformed in terms of accuracy, precision, recall, and F1-score, firmly establishing itself as a robust solution for mitigating security challenges in IoT networks.

In our future research endeavours, we will intricately focus on enhancing the adaptability of our ensemble model for different IoT networks and devices, exploring scaling possibilities and ensuring seamless integration into diverse infrastructures. This targeted effort aims to maximize the model's applicability, allowing it to effectively address the unique challenges posed by various types of IoT environments.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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